

POLITECNICO
MILANO 1863

054323 - INTERNET OF THINGS

POLITECNICO DI MILANO

COMPUTER SCIENCE AND ENGINEERING

IoT Challenge 3
Part 2

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Date: May 12, 2026

1 Exercise

1.1 Given parameters

Parameter	Value
Carrier frequency	868 MHz (Europe)
Bandwidth	125 kHz
Number of gateways	1
Number of sensor nodes	$N = 40$
Application payload size	20 bytes
LoRaWAN overhead (hdr + MIC)	13 bytes $\Rightarrow$ total packet = 33 bytes
Traffic model	Poisson, $\lambda = 2$ pkt/min per node

Table 1: System parameters

1.2 EQ1

Model: Pure (unslotted) ALOHA - transmissions are uncoordinated and may overlap.

1.2.1 Assumptions

The following assumptions are made for EQ1:

- All nodes use the **same spreading factor** SF.
- Transmissions are **uncoordinated** and may overlap in time: the system is modelled as **pure (unslotted) ALOHA**.
- The channel is ideal (no losses beyond collisions; no retransmissions).
- The full LoRaWAN packet size fed to the TTN airtime calculator is $S = 20+13 = 33$ bytes (application payload + MAC header 9 B + MIC 4 B).

1.2.2 Step 1: Packet size input for the TTN airtime calculator

The TTN airtime calculator takes the full LoRaWAN packet size = application payload + LoRaWAN overhead (MAC header 9 B + MIC 4 B = 13 B):

$$S_{\text{total}} = 20 + 13 = 33 \text{ bytes} \tag{1}$$

1.2.3 Step 2: Aggregate arrival rate

Each of the $N = 40$ nodes sends at $\lambda = 2$ pkt/min $= \frac{1}{30}$ pkt/s. The total network arrival rate is:

$$\Lambda = N \cdot \lambda = 40 \times \frac{1}{30} = \frac{4}{3} \approx 1.333 \text{ pkt/s} \tag{2}$$

1.2.4 Step 3: Offered load and success probability (Pure ALOHA)

The normalised offered load is $G = \Lambda \cdot T_{\text{air}}$. Under Poisson arrivals, the pure ALOHA packet success probability is:

$$P_s = e^{-2G} \tag{3}$$

We need $P_s \geq 0.75$, i.e.:

$$G \leq -\frac{\ln(0.75)}{2} \approx 0.1438 \tag{4}$$

1.2.5 Step 4: Airtime from TTN calculator (BW = 125 kHz, CR = 4/5, 33-byte packet)

Airtime values are read from the TTN airtime calculator with BW = 125 kHz, CR = 4/5, preamble = 8, explicit header, applied to the 33-byte full packet.

SF	T_{air} **(ms)**	$G = \Lambda \cdot T_{\text{air}}$	$P_s = e^{-2G}$	P_s **(%)**	$\geq 75\%$**?**
SF7	71.936	0.09591	0.82545	82.54%	✓
SF8	133.632	0.17818	0.70023	70.02%	×
SF9	246.784	0.32905	0.51784	51.78%	×
SF10	452.608	0.60348	0.29911	29.91%	×
SF11	987.136	1.31618	0.07191	7.19%	×
SF12	1810.432	2.41391	0.00800	0.80%	×

Table 2: Pure ALOHA success probability for each SF (33-byte LoRaWAN packet, BW = 125 kHz). Low-DR optimisation active for SF11 and SF12.

Result EQ1

The largest spreading factor that guarantees $P_s \geq 75\%$ is **SF7** ($P_s = 82.54\%$). SF8 and above yield a success rate below 75%.

1.3 EQ2

1.3.1 Assumptions

After deploying the network in the field, two symptoms are observed:

- The measured packet success rate is **significantly lower** than the theoretical prediction.
- Performance is **non-uniform across nodes**: some nodes achieve relatively good delivery rates, while others experience very poor performance.

The three candidate actions are:

1. Decrease the number of nodes
2. Move the nodes closer to the gateway
3. Increase the spreading factor

The key symptom is **non-uniform performance**: some nodes deliver well, others poorly. This asymmetry is the hallmark of a **physical-layer / link-budget problem**, not a collision problem (collisions affect all nodes roughly equally).

1.3.2 Analysis of each option

Option	Justification
1. Decrease nodes	Fewer nodes lower G and reduce collisions, improving average P_s uniformly. This would not explain why some nodes perform well and others do not: collision-induced losses are symmetric across all nodes. Reducing the number of nodes is also undesirable if full coverage is needed.
2. Move nodes closer	Nodes far from the gateway experience higher path loss, lower SNR and therefore high frame-error-rate due to channel impairments, independently of collisions. Close nodes decode reliably; distant ones fail frequently. This produces exactly the observed non-uniform pattern. Reducing the distance improves the link budget for the worst-performing nodes and addresses the root cause.
3. Increase SF	A higher SF provides better link budget (sensitivity) but also increases T_{air}, raising G and lowering P_s (EQ1 already shows SF8 falls below 75%). More importantly, a single global SF increase does not inherently fix the asymmetry: distant nodes still see a worse SNR than close ones. ADR (Adaptive Data Rate) could assign different SFs per node, but that is not what this option asks.

Table 3: Justification for each candidate action

Result EQ2

The most appropriate action is **option 2 - Move the nodes closer to the gateway**. The non-uniform delivery rates indicate a link-quality problem: distant nodes receive a weak signal and suffer high packet error rate, while close nodes work well. Improving the link budget (shorter distance) addresses the root cause of the asymmetric performance.